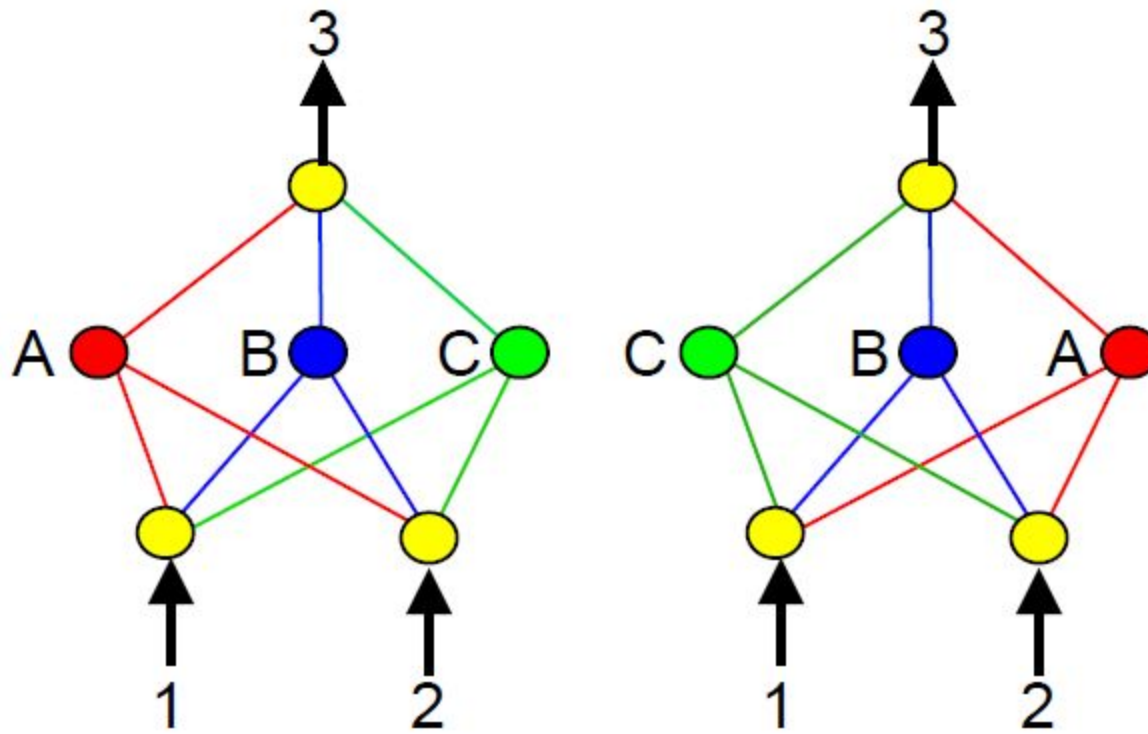


NEAT / HyperNEAT



$$\begin{array}{r} [A,B,C] \\ \times [C,B,A] \\ \hline \end{array}$$

Crossovers: $[A,B,A]$ $[C,B,C]$
 (both are missing information)

Competing Conventions:

Two neural networks:

Both encode the same function;

Have different conventions for doing so.

No matter how they're crossed, their children will lack information.

Stanley, K.O., Miikkulainen
 (2001) Evolving Neural
 Networks through
 Augmenting Topologies.

NEAT / HyperNEAT

Genome (Genotype)

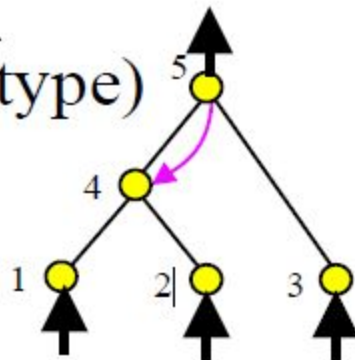
Node
Genes

Node 1	Node 2	Node 3	Node 4	Node 5
Sensor	Sensor	Sensor	Hidden	Output

Connect.
Genes

In 1	In 2	In 2	In 3	In 4	In 5
Out 4	Out 4	Out 5	Out 5	Out 5	Out 4
Weight 0.7	Weight -0.5	Weight 0.5	Weight 0.2	Weight 0.4	Weight 0.6
Enabled	Enabled	DISABLED	Enabled	Enabled	Enabled
Innov 1	Innov 3	Innov 4	Innov 5	Innov 6	Innov 10

Network (Phenotype)



Genetic encoding of
neural networks
in (N)euro(e)volution
of (A)ugmenting
(T)opologies: (NEAT)

Stanley, K.O., Miikkulainen
(2001) Evolving Neural
Networks through
Augmenting Topologies.

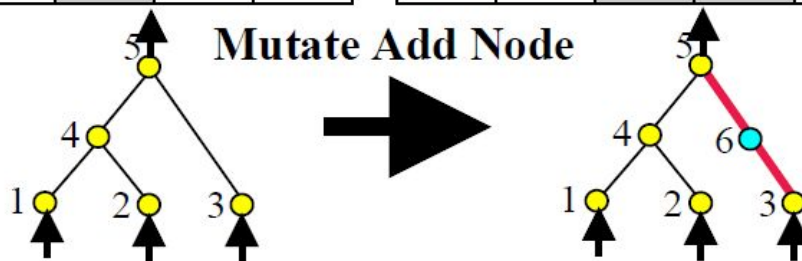
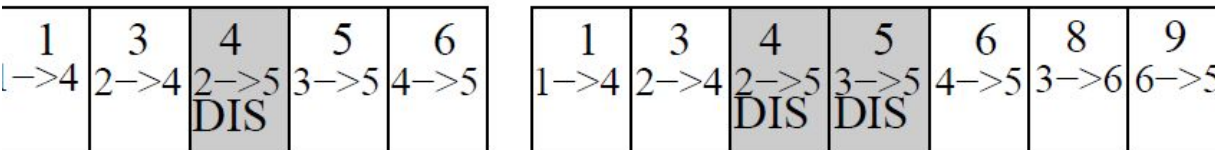
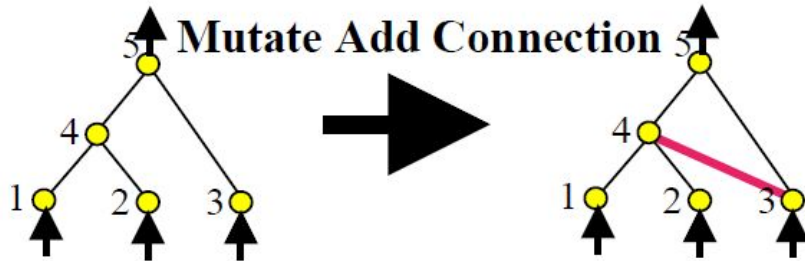
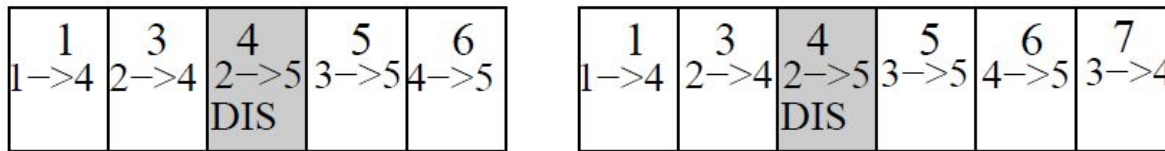
NEAT / HyperNEAT

Historical Markings:

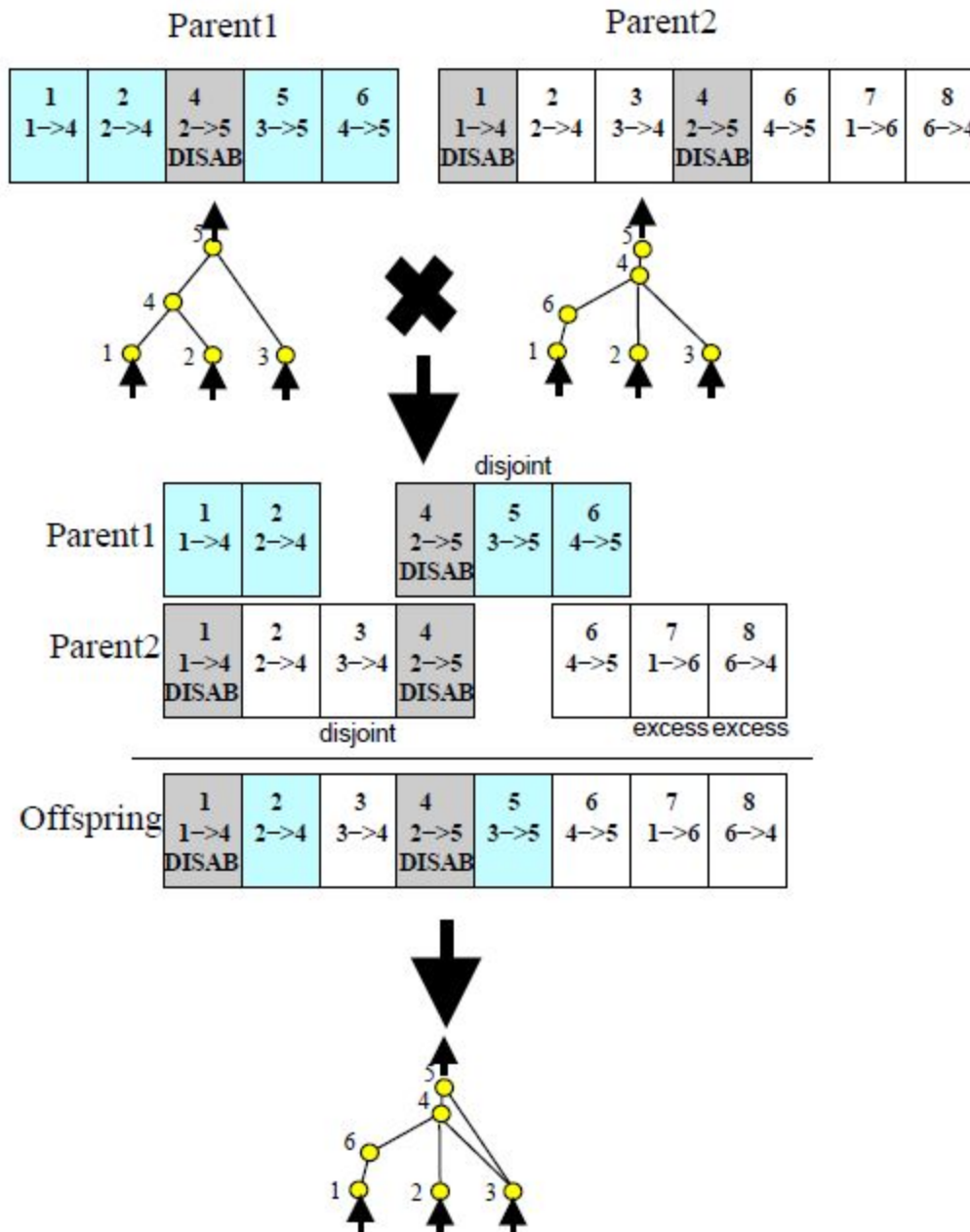
Keep a global counter; every time a neuron or synapse is added, assign the value of the counter, and increment it.

Genes/synapses can be disabled, but remain in the genome.

Stanley, K.O., Miikkulainen
(2001) Evolving Neural
Networks through
Augmenting Topologies.



NEAT / HyperNEAT



NEAT designed so
That crossover is

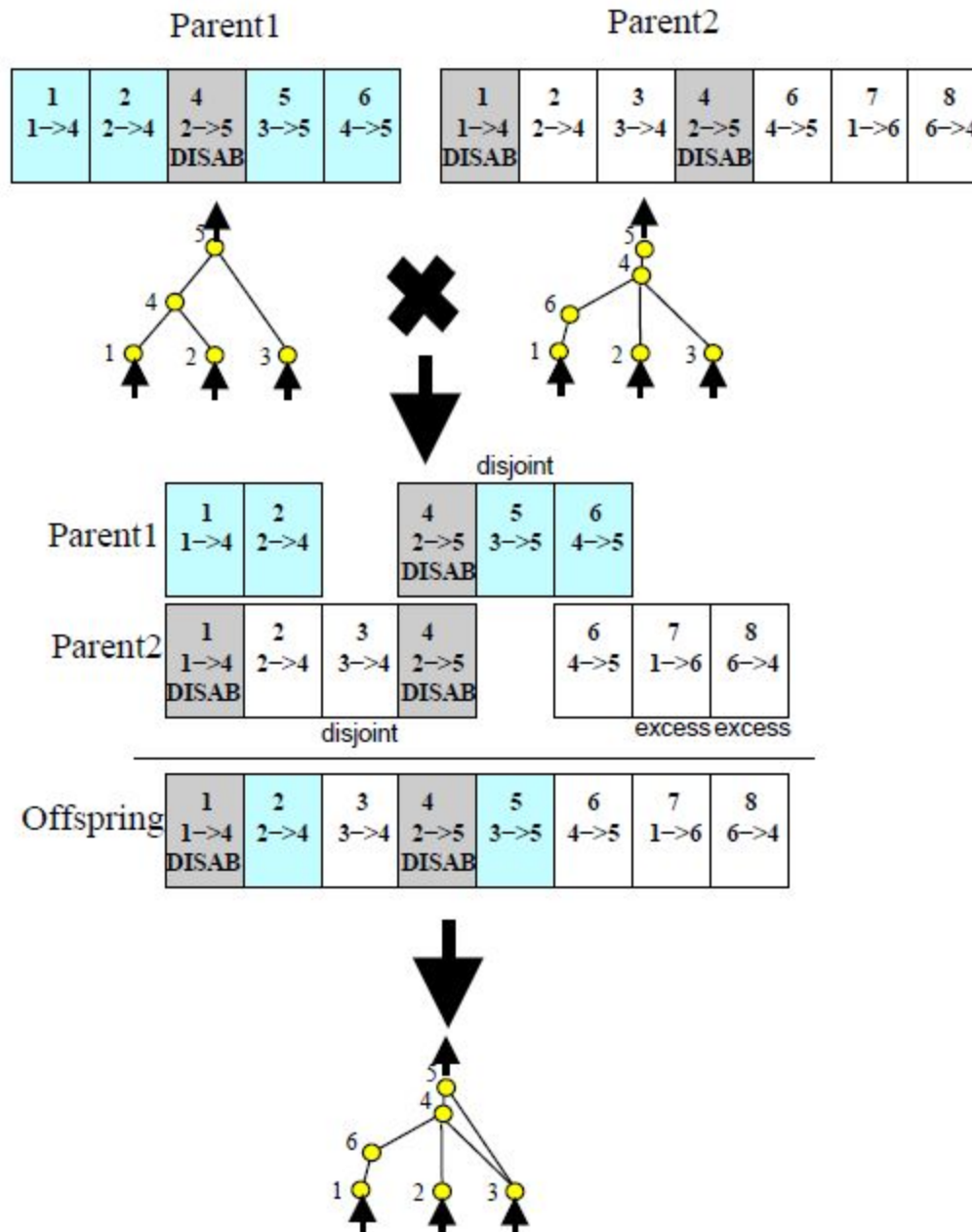
(1) Algorithmically simple

and

(2) Produces children
that are similar to their
parents.

Stanley, K.O., Miikkulainen
(2001) Evolving Neural
Networks through
Augmenting Topologies.

NEAT / HyperNEAT



Take two parent
NNs:

Line up connection genes
according to their
historical markings.

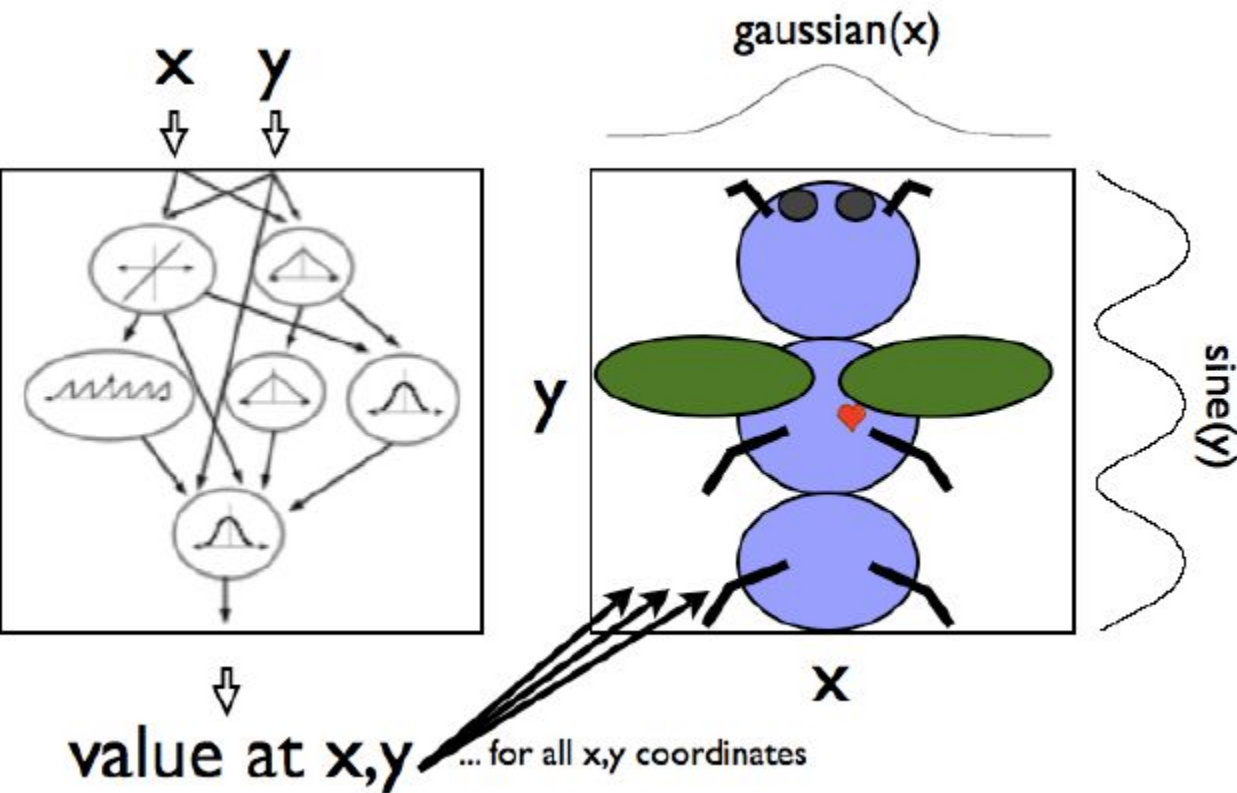
For matching genes,
copy either gene into child
at random.

Disjoint genes (those
in the middle without a
partner gene)

Excess genes (those at
the end)

Stanley, K.O., Miikkulainen
(2001) Evolving Neural
Networks through
Augmenting Topologies.

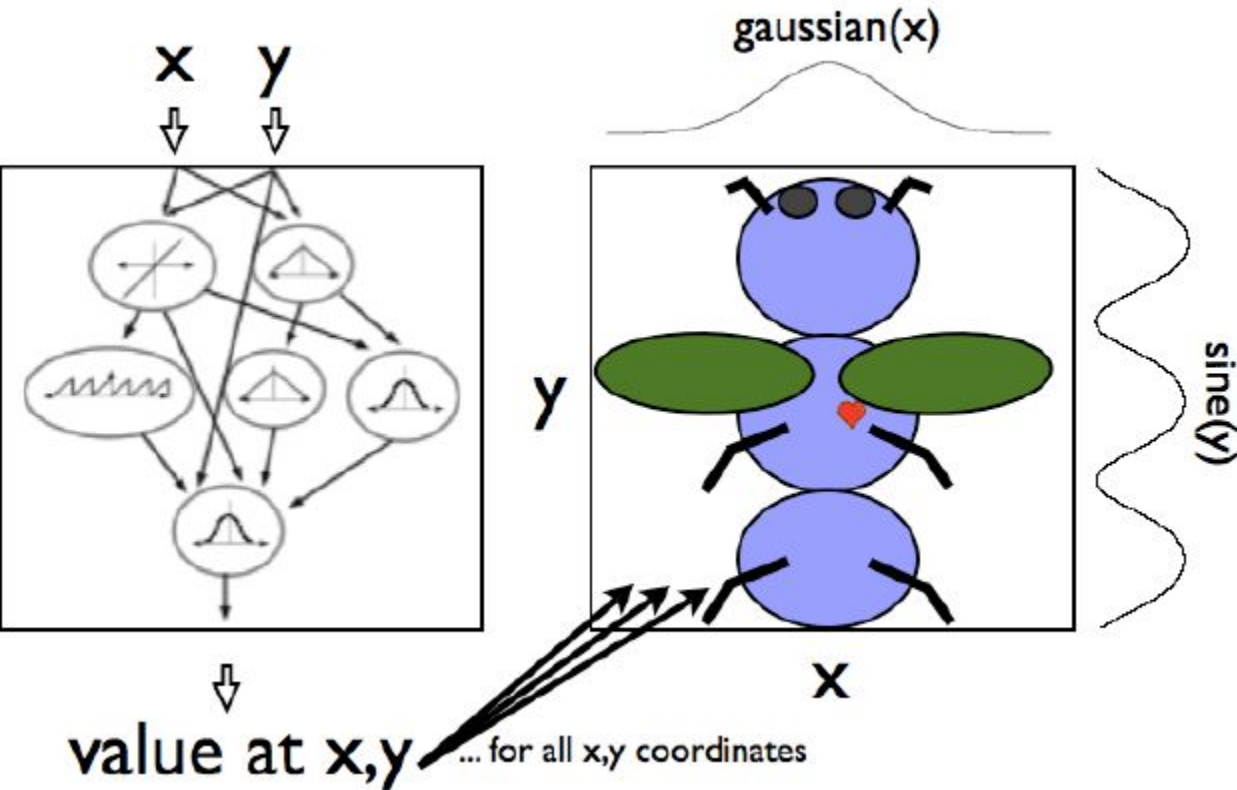
NEAT / HyperNEAT



HyperNEAT:

Evolves neural networks
(compositional
pattern-producing
networks)
that produce regular
patterns spatially:

NEAT / HyperNEAT

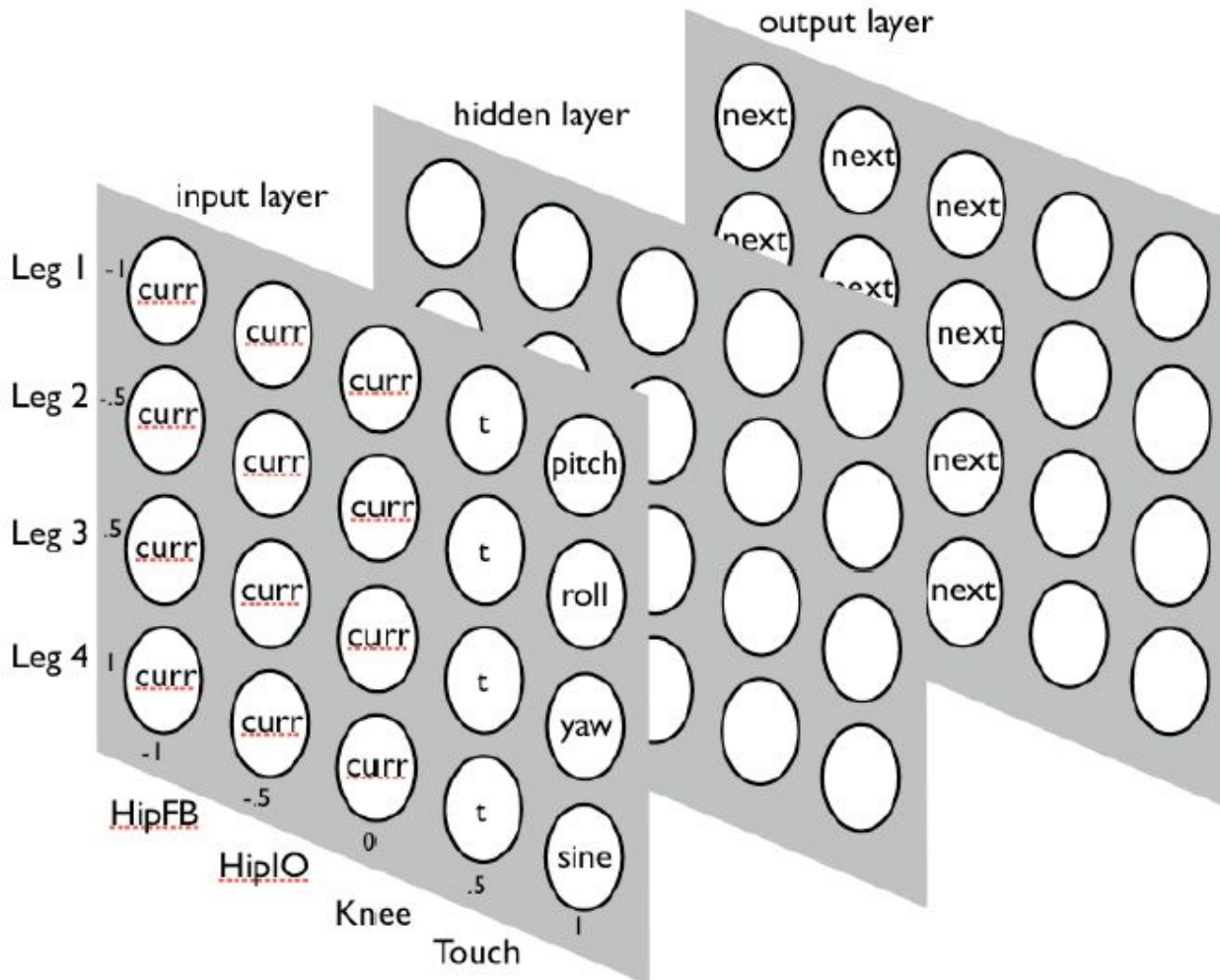


HyperNEAT:

HyperNEAT "paints" regular patterns on to a hypercube.

The dimensionality of the hypercube is determined by the dimension of the input coordinates.

NEAT / HyperNEAT



HyperNEAT can
be used
to “paint” weights
on to synapses of
a second neural network.

...why do this,
if NEAT already evolves
neural networks?

NEAT / HyperNEAT

HyperNEAT



FT-NEAT



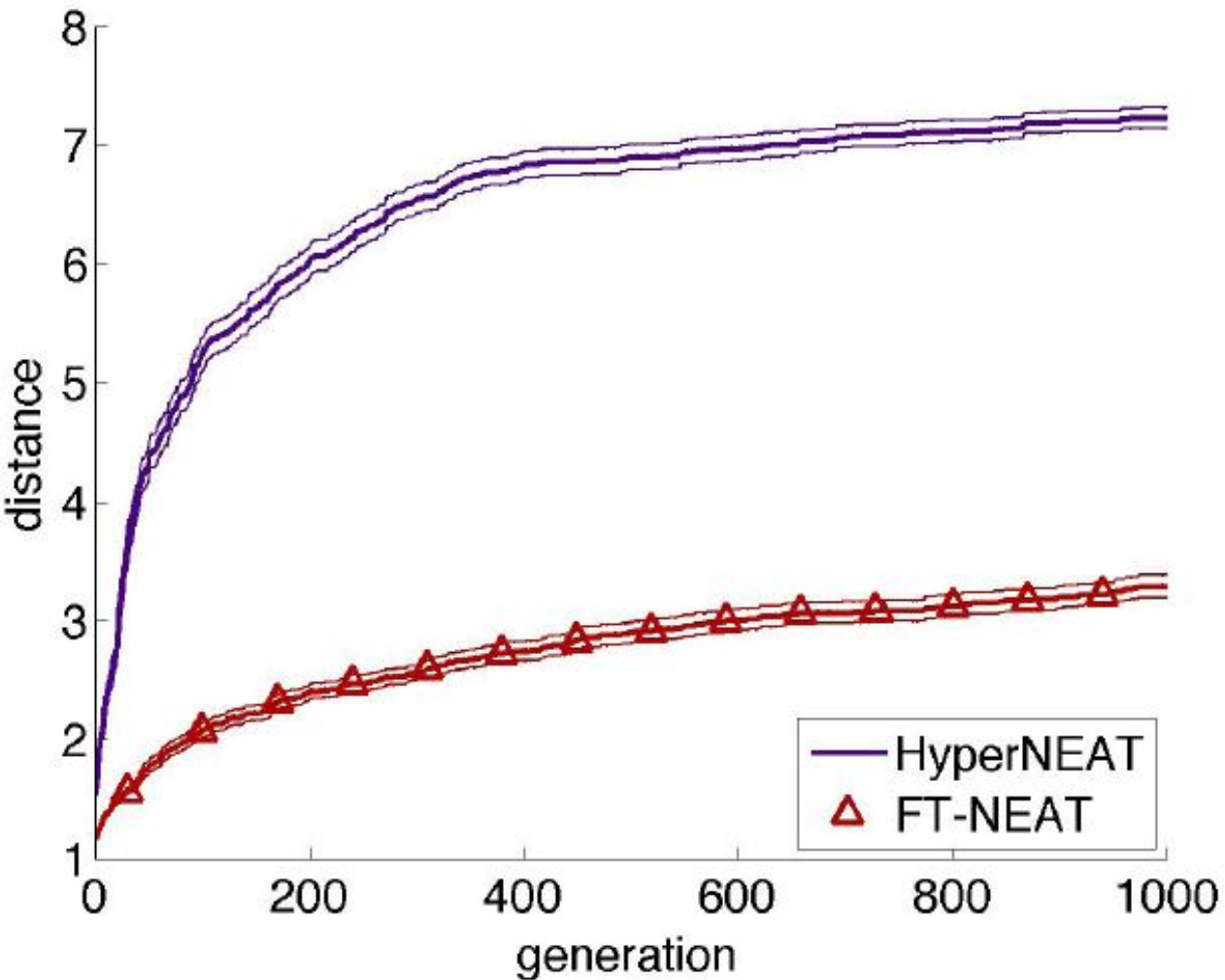
Compare HyperNEAT to NEAT:

FT-NEAT: Fixed Topology NEAT.

Use same NN as in HyperNEAT;
allow for mutation and crossover, but
not the addition/removal of neurons/synapses.

Clune, J. *et al.* (2009)
Evolving Coordinated
Quadruped Gaits with the
HyperNEAT Generative
Encoding

NEAT / HyperNEAT

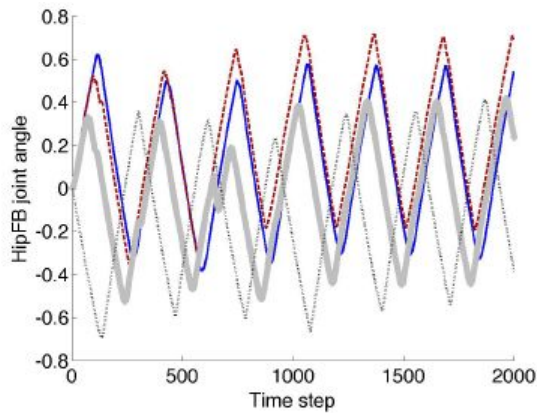
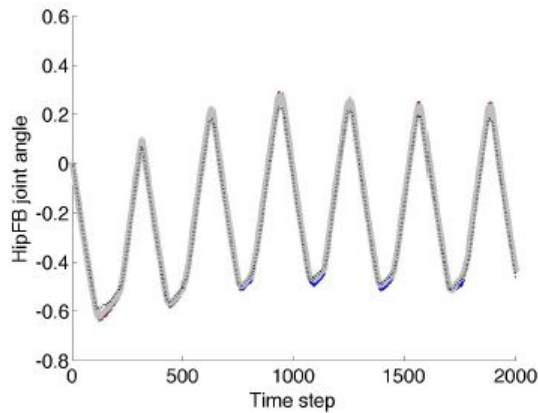


Results from
evolving locomotion.

Clune, J. *et al.* (2009)
Evolving Coordinated
Quadruped Gaits with the
HyperNEAT Generative
Encoding

NEAT / HyperNEAT

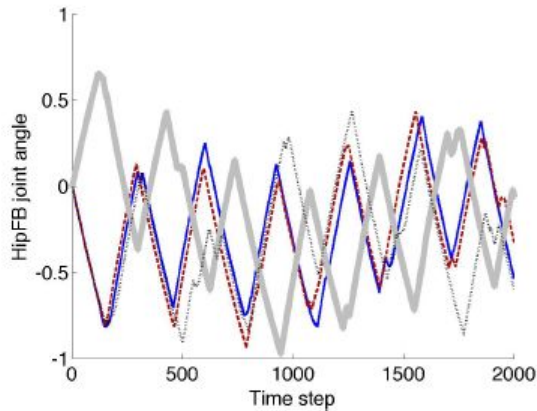
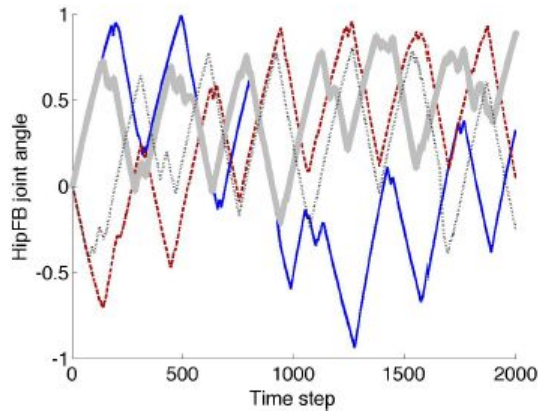
HyperNEAT



HyperNEAT repeatedly finds regular gaits;

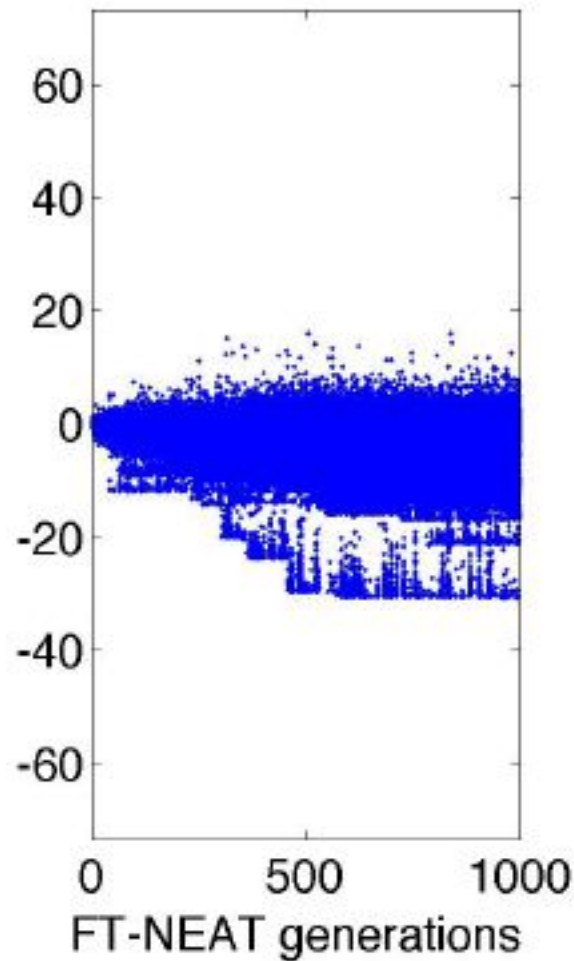
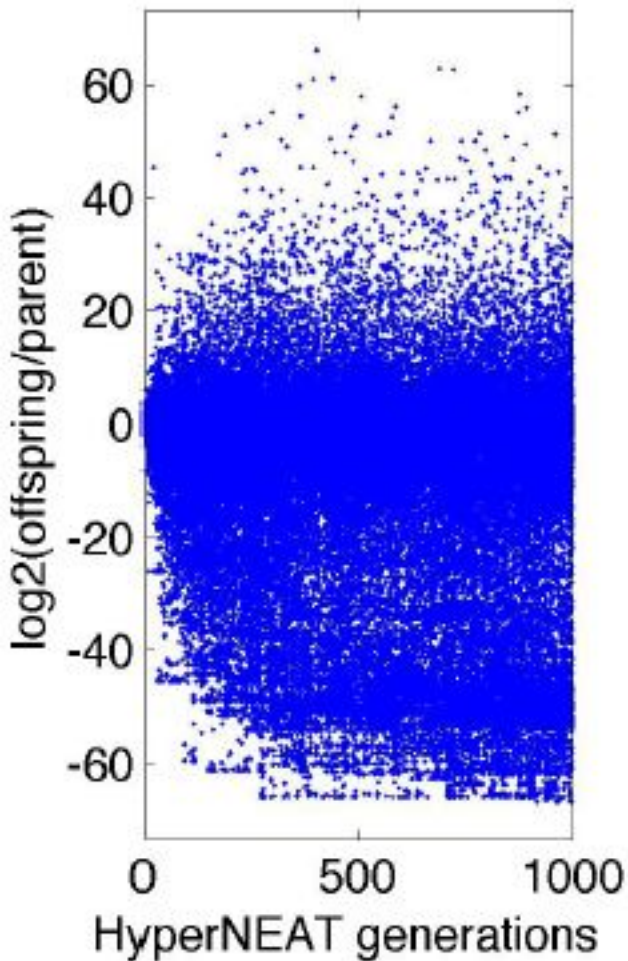
FT-NEAT does not.

FT-NEAT



Clune, J. *et al.* (2009)
Evolving Coordinated
Quadruped Gaits with the
HyperNEAT Generative
Encoding

NEAT / HyperNEAT



For mutations,

mean fitness of children
compared to parents:

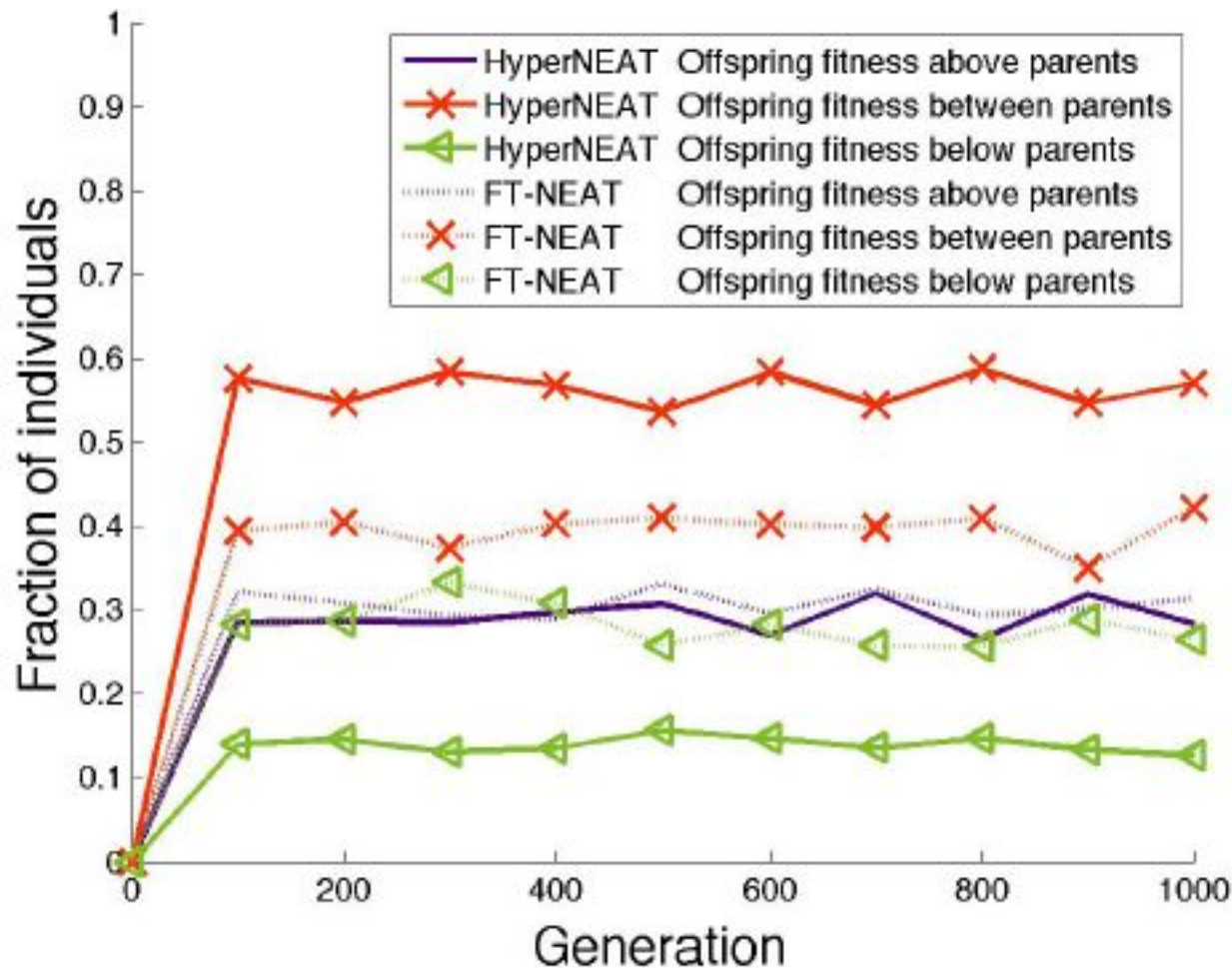
HyperNEAT:
FT-NEAT:

but HyperNEAT tends
to create more fit
children than FT-NEAT.

..why?

It also creates much
worse children, but
these are discarded.

NEAT / HyperNEAT



If children are produced by crossover...

In HyperNEAT, offspring tend to be more like their parents than in FT-NEAT.

...why?